POE PART 1- Planning & Design

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I hereby declare that I did not plagiarize the content of this assignment and that this is my own work.

Assignment submitted via Turnitin: ✓ (Tick the Box)

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Planning & Design

Introduction

The following document will discuss the planning and design concepts of our budget app, MoneyCane. It will demonstrate the build up from our research report including elements extracted from the strengths and innovative features researched from Spendee, MyMoney and Cashew. This document demonstrates the elaboration of our apps overview, icon choice, features, requirements and project plan in the form of a Gantt chart.

Overview

MoneyCane is a budgeting app designed to help users track their income, expenses, and shared finances. It improves financial awareness by providing spending insights, smart alerts, and visual analysis to help users understand their spending habits.

The app is designed for individuals and groups who want to manage money together. Users can explore the app using demo mode or continue without signing in for offline use; however, signing in allows users to securely back up their data and access additional features.

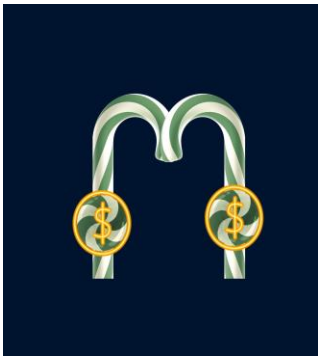


Figure 1:MoneyCane Icon

The icon features 2 sugarcanes forming the letter “M” to represent the app name. The gold coins represent income and expenses, while the green color wrapped around the sugarcanes represents money.

Innovative Features:

- Shared wallet for managing group expenses (Lahiri, 2026)
- Spending overview dashboard (Kokosha, 2025)
- Smart alerts for overspending budget limits (Lahiri, 2026)
- Expense and income analysis (Pareek, 2022)
- Secure pin authentication (Lahiri, 2026)
- Demo mode (Kokosha, 2025)
- Upload documents or receipts when making entries (optional) (Weber, 2025)
- Badge gamification for Rewards(Chacon, 2024)
- Light mode, dark mode and home screen customization (Kokosha, 2025)
- Data backup (Kokosha, 2025)
- Continue without signing in for offline use (Weber, 2025)

List of detailed requirements

Register and Login Screen:

- User is required to register and log in into MoneyCane using the username and password they assigned themselves during registration (Kokosha, 2025).
- Users will be able to view the demo of MoneyCane before registering. Demo will already be populated with dummy data, but user can edit the demo to test functionality (Kokosha, 2025).
- Users can use MoneyCane offline by clicking the "continue without signing in" button in the registration screen but will need to sign in to back up data and access online features (Kokosha, 2025).
- Users have the option to sign in with google to complete registration this is done through an API to authenticate the user (Lahiri, 2026)

Database and Backing up data:

- Users should be able to back up their data on google drive (Kokosha, 2025).
- Make use of a local database to save all the users financial data (Pareek, 2022).

General features:

- The user is required to set a PIN for security which will be prompted every time they enter MoneyCane (Lahiri, 2022).
- Navigation bar/menu has the following: Home, Budget, Analysis, Rewards and More screens/icons (Weber, 2025).
- Users can choose if they want light or dark mode (Kokosha, 2025).
-

User Wallets:

- There should be 1 wallet initially; the user needs to enter the amount of money available in that wallet (Weber, 2025).
- Users will be able to add more types of wallets by adding the name and amount (Weber, 2025).

- Wallets are like the types of accounts we get, such as cash, bank cards, savings, etc. (Lahiri, 2026).
- When users enter their income transactions, they can choose the type of income it is. Income categories include salary, business, extra income, etc. (Lahiri, 2026).
- Users can add other types of income categories (Lahiri, 2026).
- Incomes entered can also be listed as a recurring income, so it can be repeated each month like a salary (Lahiri, 2026).
- When the user enters an income transaction, they will be able to choose which wallet the income is being added to (Weber, 2025).

Entering Expenses:

- User should be able to add an expense entry; entry includes the date, description, amount, start and end times and category (Pareek, 2022).
- User can voluntarily add a picture of proof of payment/reference/receipt to every expense entry (Kokosha, 2025).
- Users can choose a category from the list provided on MoneyCane. Categories provided consist of everyday expenses such as groceries, shopping, transport, healthcare etc. (Lahiri, 2026)
- User has the option to manually create their own category; in the case they do not find a category option they want during an expense entry (Kokosha, 2025).
- User must also state which wallet the expense is being taken from (Kokosha, 2025).
- User can select if an expense will recur every week/month/year- like a subscription. Therefore, the expense should automatically be repeated by MoneyCane on the chosen date (Lahiri, 2026).
- The start and end date of an expense can be included (optional) (Lahiri, 2026).
- The date when an expense is entered will be set to the current date, but the user should be able to change the date to the past in case they forgot to enter an expense that day (Pareek, 2022).

Budgeting Limits:

- User has the ability to set a spending limit for categories respectively per month which the user will be able to monitor visually with the help of a progress dashboard (Kokosha, 2025).

- If the user is over their category limit or monthly budget, then it will be highlighted on the dashboard (Kokosha, 2025).
- Users can set a monthly spending budget goal; they will be able to track their progress with the use of a progress dashboard displayed on the home screen (Kokosha, 2025).
- Furthermore, users are required to set a minimum and maximum monthly goal regarding the amount of money they are set on spending (Kokosha, 2025).
- When the user goes over the budget then they will get a notification on the home screen. After they click it, they will be directed to the budget screen where the budget widget progress bar will be highlighted (Lahiri, 2022).

Home screen features:

- Home screen should display the following features: wallet balance, graph displaying monthly budget left, progress bar/dashboard, and recent transactions (Kokosha, 2025).
- Home screen should include the spending overview or summary from the start of the month to date in the form of a line graph (Weber, 2025)
- The progress bar is determined by the budgets set (Kokosha, 2025).
- Users can choose which of the above features they would like to see. Initially, everything will be displayed, but the user can choose the information they would like to display (Kokosha, 2025).
- User is able to set a goal for the week in the home/rewards screen. The progress of their goal will be displayed in the Rewards screen (Chacon, 2024).

More Screen features:

- This screen contains more actions the user needs.
- It should contain the following functions/widgets/buttons for, but is not limited to: User Profile, Settings, Notifications, Calendae, Goals, Subscription, Wallets, Budgets, Categories, Transactions, Login, Feedback, Shared Wallets, and Spending Summary (Kokosha, 2025).
- The above should act as hyperlinks to another screen that is providing that functionality (Kokosha, 2025).

Analysis Screen features:

- Users should be capable of viewing a list of their transactions over a specific period chosen. User can click on the transaction and see a detailed description of the expense entry, including retrieving the image attached (Pareek, 2022).
- Additionally, users have the potential to view the total amount spent on each category throughout a specific time selected (Pareek, 2022).
- A graph displaying users daily spending over chosen timespan that includes the category and amount enabling the user to spot patterns. The graph should also show the minimum and maximum goals set by the user (Kokosha, 2025).
- Users should be able to track how well they are progressing in staying within their spending goals in a visual format such as a pie chart or line graph (Lahiri, 2026)
- The user should be able to see which category took up most of the budget; this can be displayed in a pie chart, and this information is updated weekly (Kokosha, 2025).
- User will see a calendar and can click a day to see a summary of the transactions that took place (Weber, 2025)

Gamification (Rewards) feature:

- At the end of the month, provided that the user stayed within their budget limit for a category or the month, then they will win a badge in the form of a green candy cane (Chacon, 2024).
- A notification should be sent when the cane has been filled 100% (Lahiri, 2026). The cane functions as a progress bar (Chacon, 2024).
- The bottom corner of the screen will have a cane icon displaying the number of canes they have won (Chacon, 2024).

Shared Wallet feature:

- Allows multiple users to manage and track finances within one shared wallet (Lahiri, 2026).
- User 1 will invite user 2 to join a shared wallet via email. User 2 will have to accept or decline the invite in their MoneyCane account (Lahiri, 2026).
- If accepted, then users must be able to manage transactions in the shared wallet (Lahiri, 2026).
- The email invitation process is done using an API and relationships between database tables to assist in linking users to the same wallet (Lahiri, 2026).

Automation:

- Smart Alert: Users will get reminders to enter their expenses and an alert if they are reaching their budget/category limit via notifications (Lahiri, 2026)
- After the user sets a monthly budget limit, the widget will give a suggestion on how the user can spend their money to stay within budget, for example: “Spend R350/day to stay within budget”. App calculates this after the user sets a budget limit (Kokosha, 2025).
- MoneyCane must have the proficiency to handle user input errors without crashing.
- MoneyCane is expected to work on most devices through automated testing via GitHub actions (Kokosha, 2025).

User Interface Design

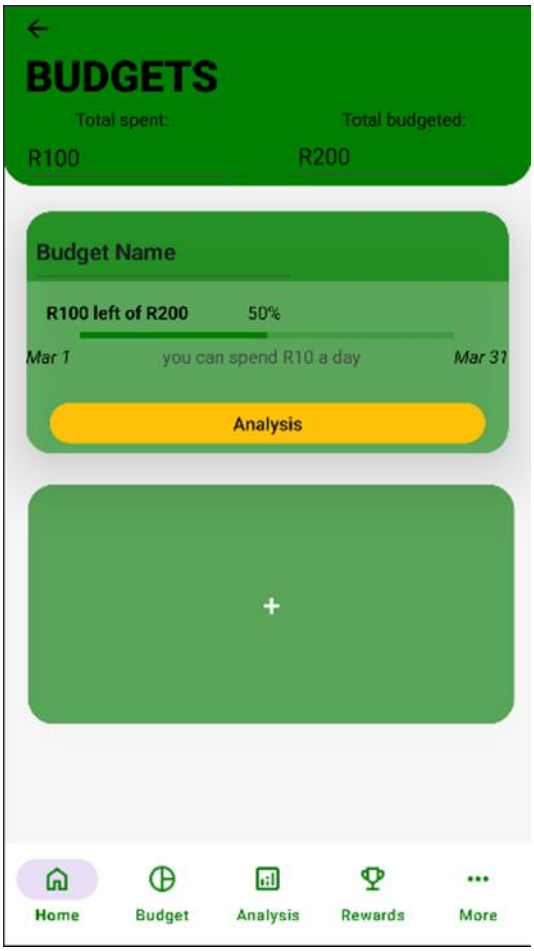


Figure 2:Budget Screen Light Mode

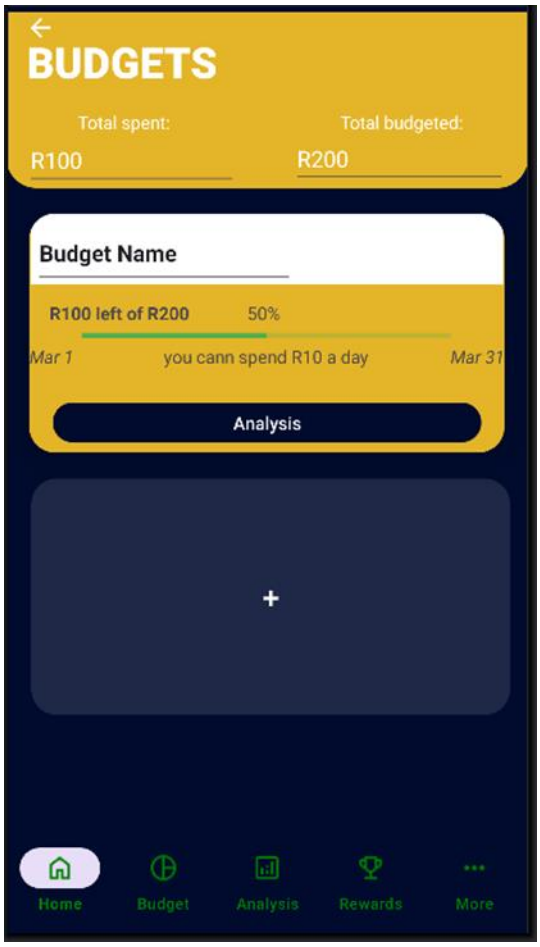


Figure 3:Budget Screen Dark Mode

This screen serves as the budget managing screen, where users can create and view all budgets they create, as well as hyperlinking to an analysis on each budget. User receives suggestions on budget spending through automation.



Figure 4:Analysis Screen Light Mode Part 1

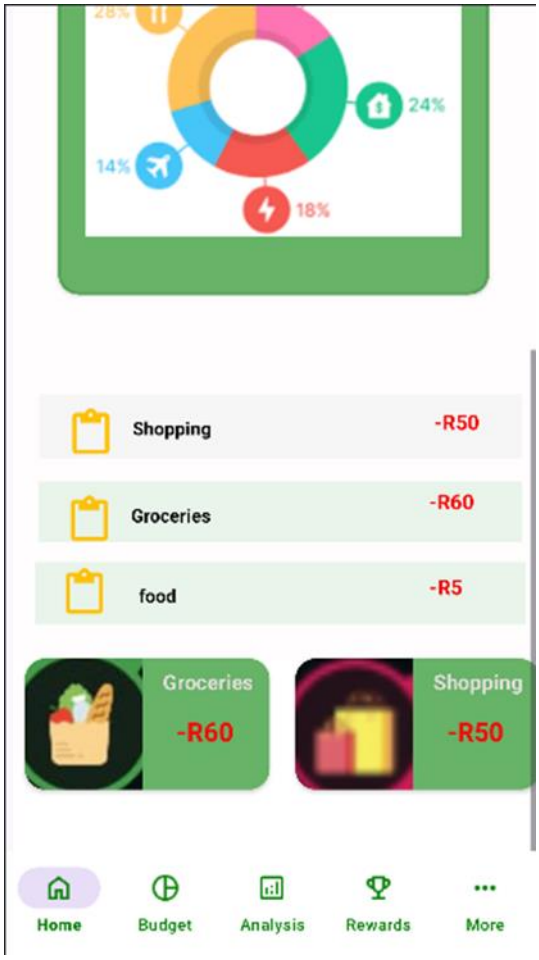


Figure 5:Analysis Screen Light Mode Part 2



Figure 6: Analysis Screen Dark Mode Part 1



Figure 7: Analysis Screen Dark Mode Part 2

Figures 4-7 are the analysis screens, they are used by the user to visualize and analyse their budgets and spending, allowing a break down on income and expenses and the option to filter the time period of the analysis. The Images have been split due to the scroll view emphasizing how detailed the analysis screen is.

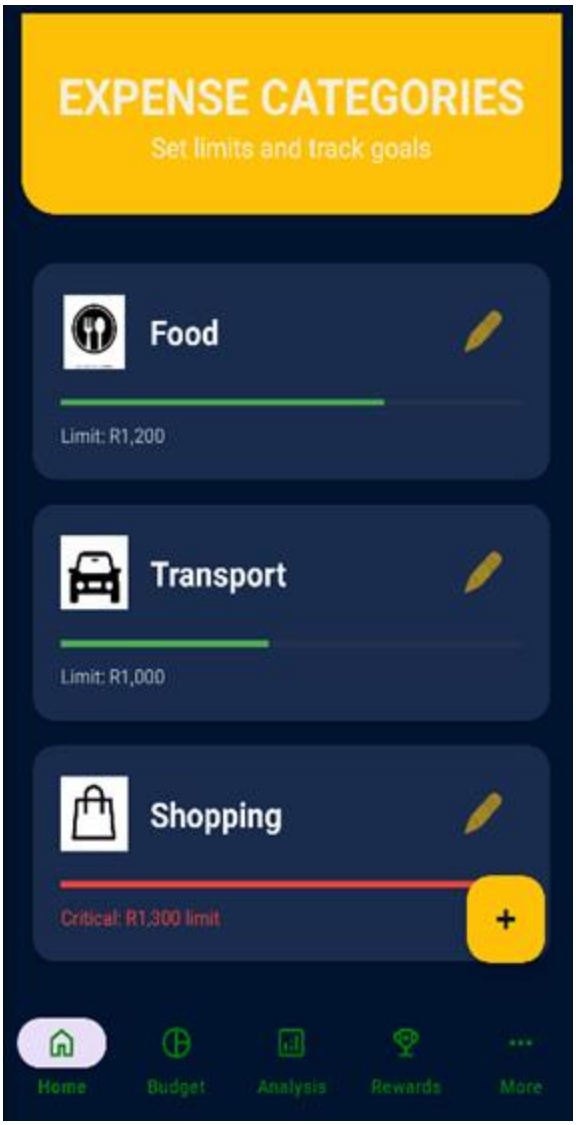


Figure 8: Category Screen Dark Mode

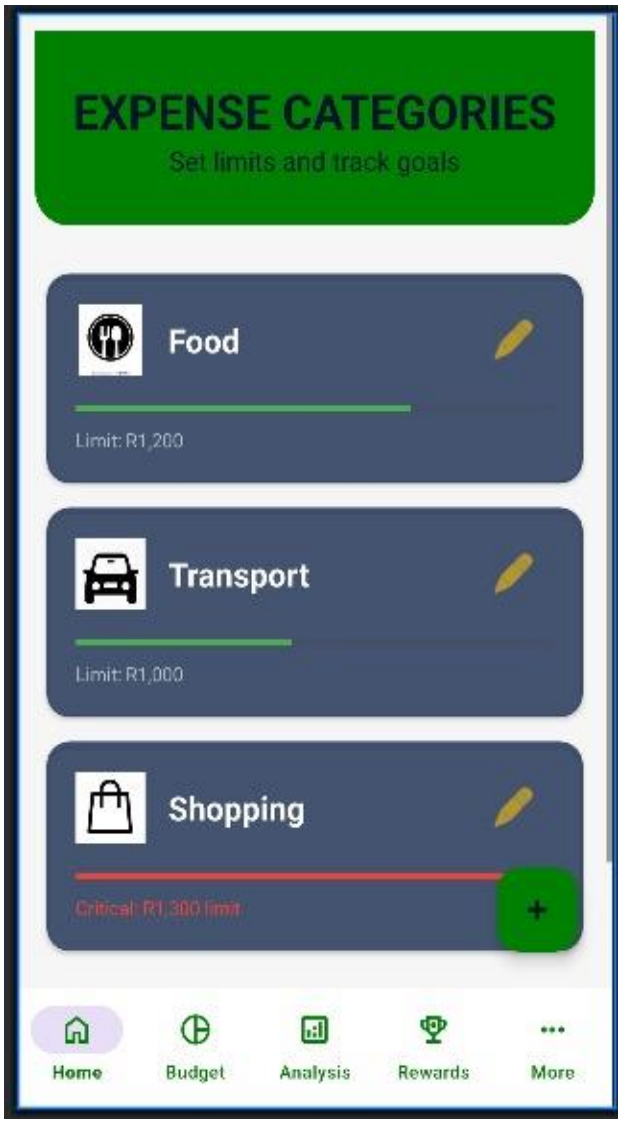


Figure 9: Category Screen Light Mode

Figures 8 & 9 serve as a budget tracking dashboard where users can monitor their spending against set limits as well as create new category limits. The progress bar helps the user visualize how much they have left to spend.

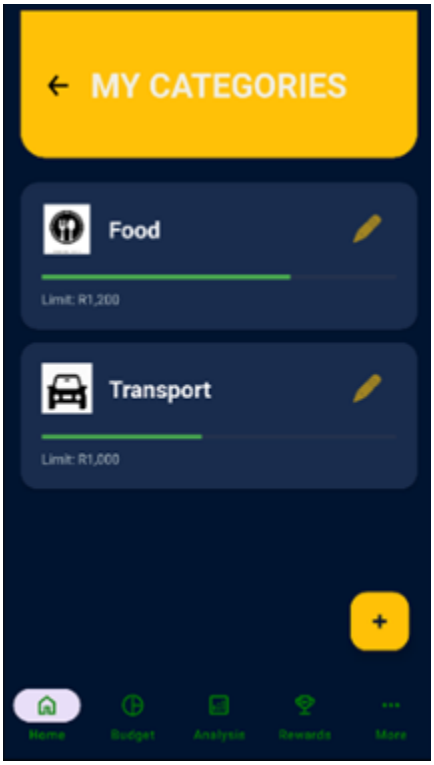


Figure 10: Personalized Category Screen Dark Mode

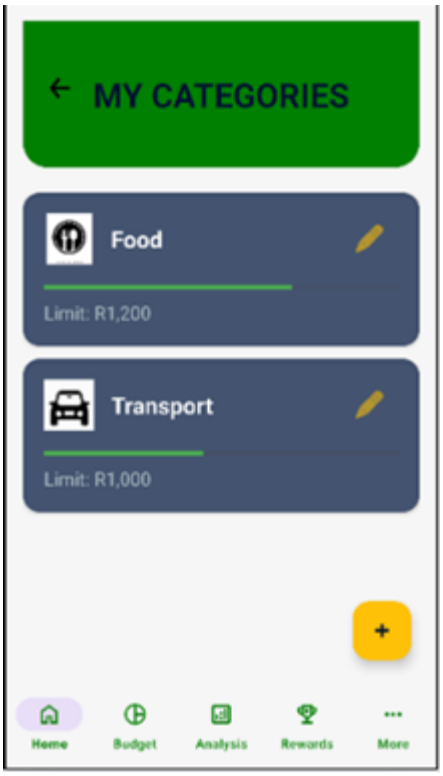


Figure 11: Personalized Category Screen Light Mode

Displays a personalised list of spending categories, allowing users to manage their budgets as well as editing the expenses. This customization allows the user to not be limited to set categories but create new spending milestones for themselves.

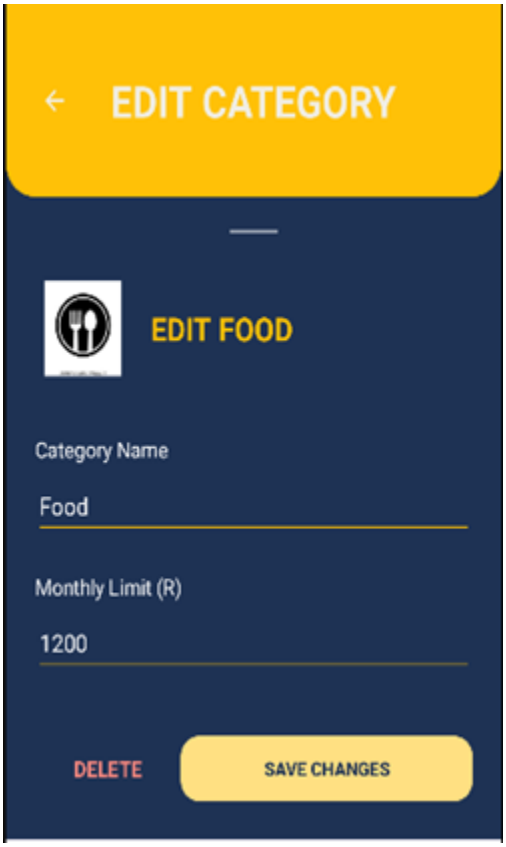


Figure 12: Edit Category Dark Mode Screen

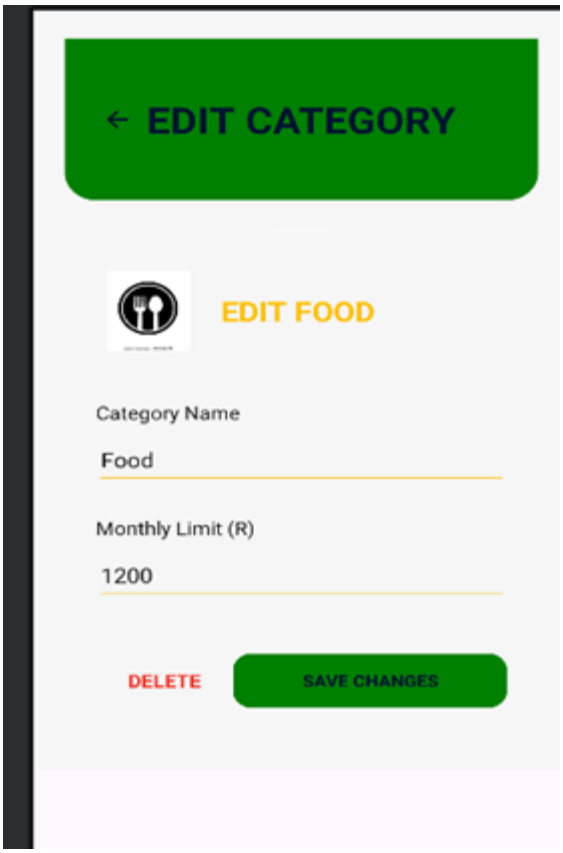


Figure 13: Edit Category Light Mode Screen

This screen allows users to modify a specific budget category by updating its name and monthly limit or by deleting the category.

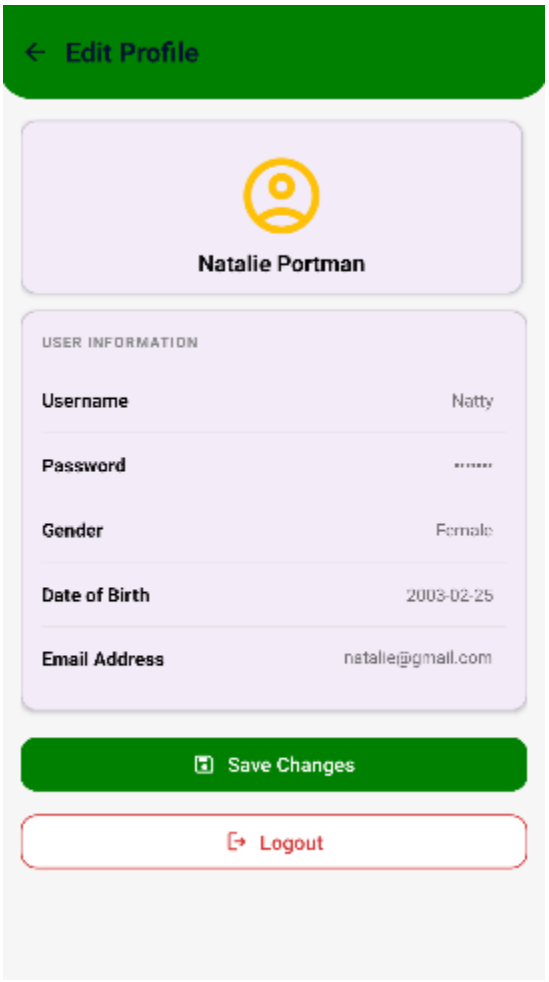


Figure 14: Edit Profile Screen Light Mode

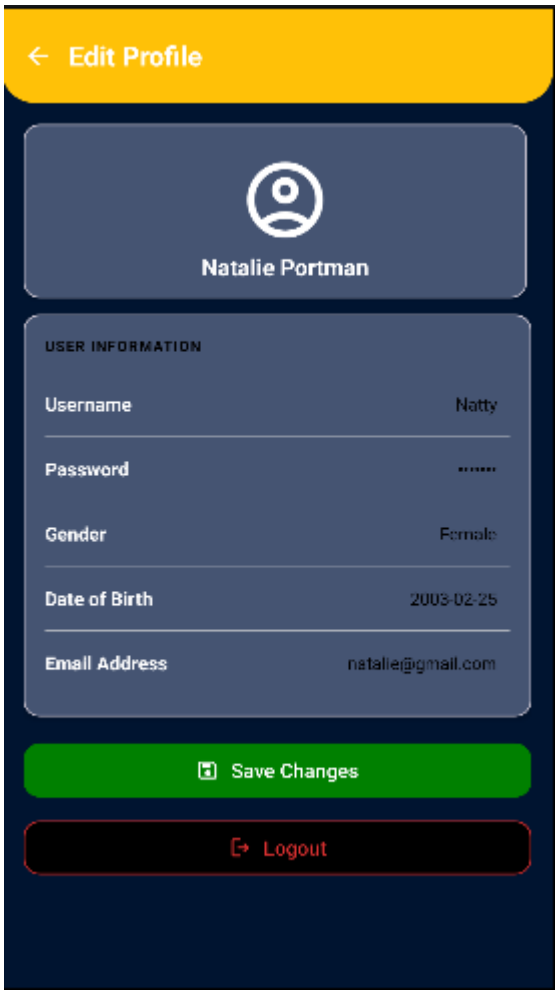


Figure 15: Edit Profile Screen Dark Mode

Figures are the edit profile screens. The user can edit their personal information and save the changes to be saved in the database. User can also logout of their MoneyCane account.

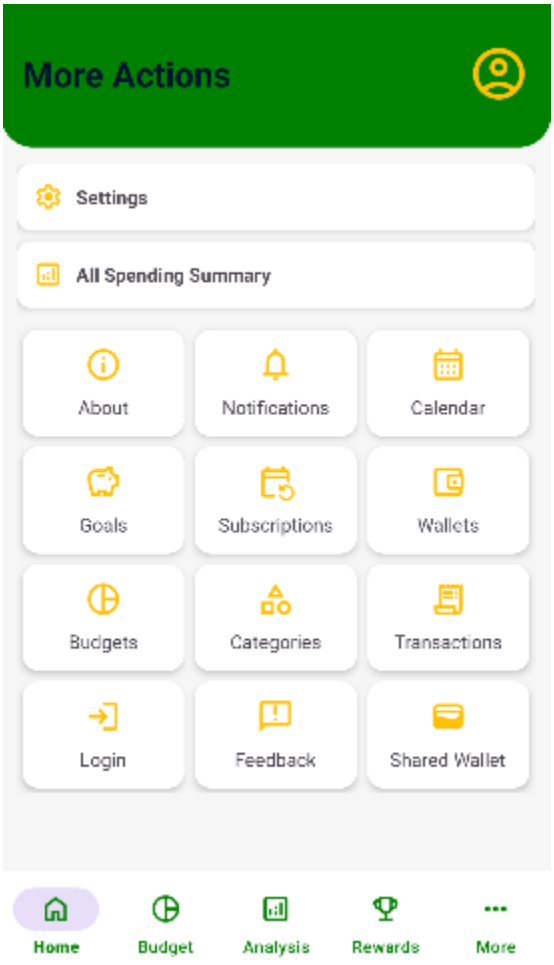


Figure 16: More Screen Light Mode

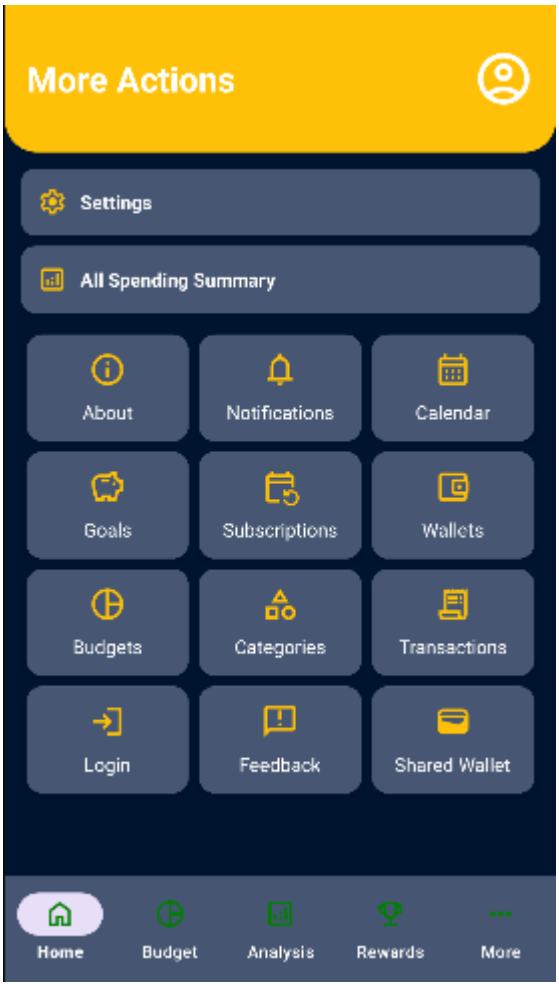


Figure 17: More Screen Dark Mode

The More screen displays all the actions that are available to the user. Each widget hyperlinks to screen performing that function. The account icon at the top corner can be clicked for the user to be redirected to the edit profile screen

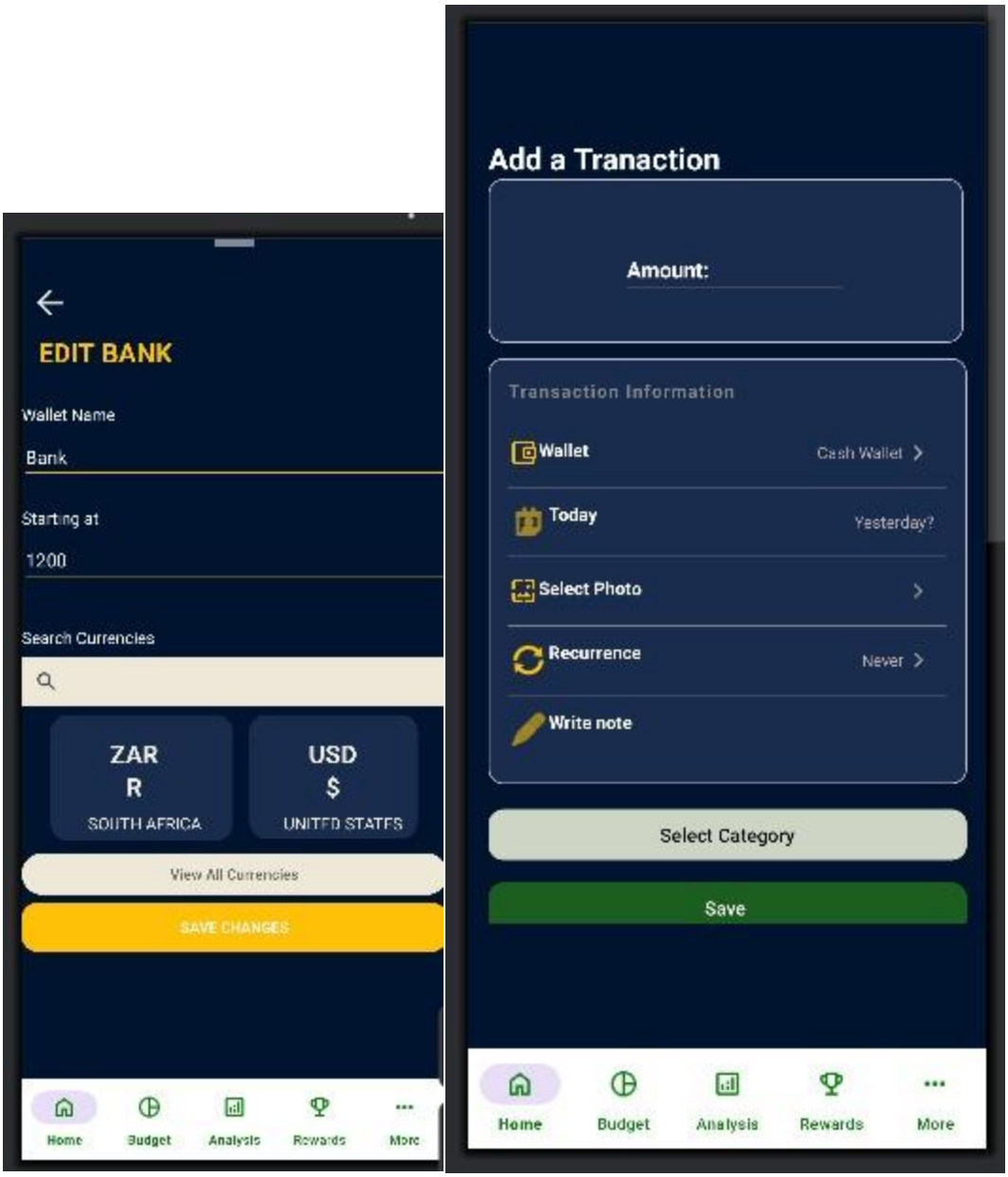


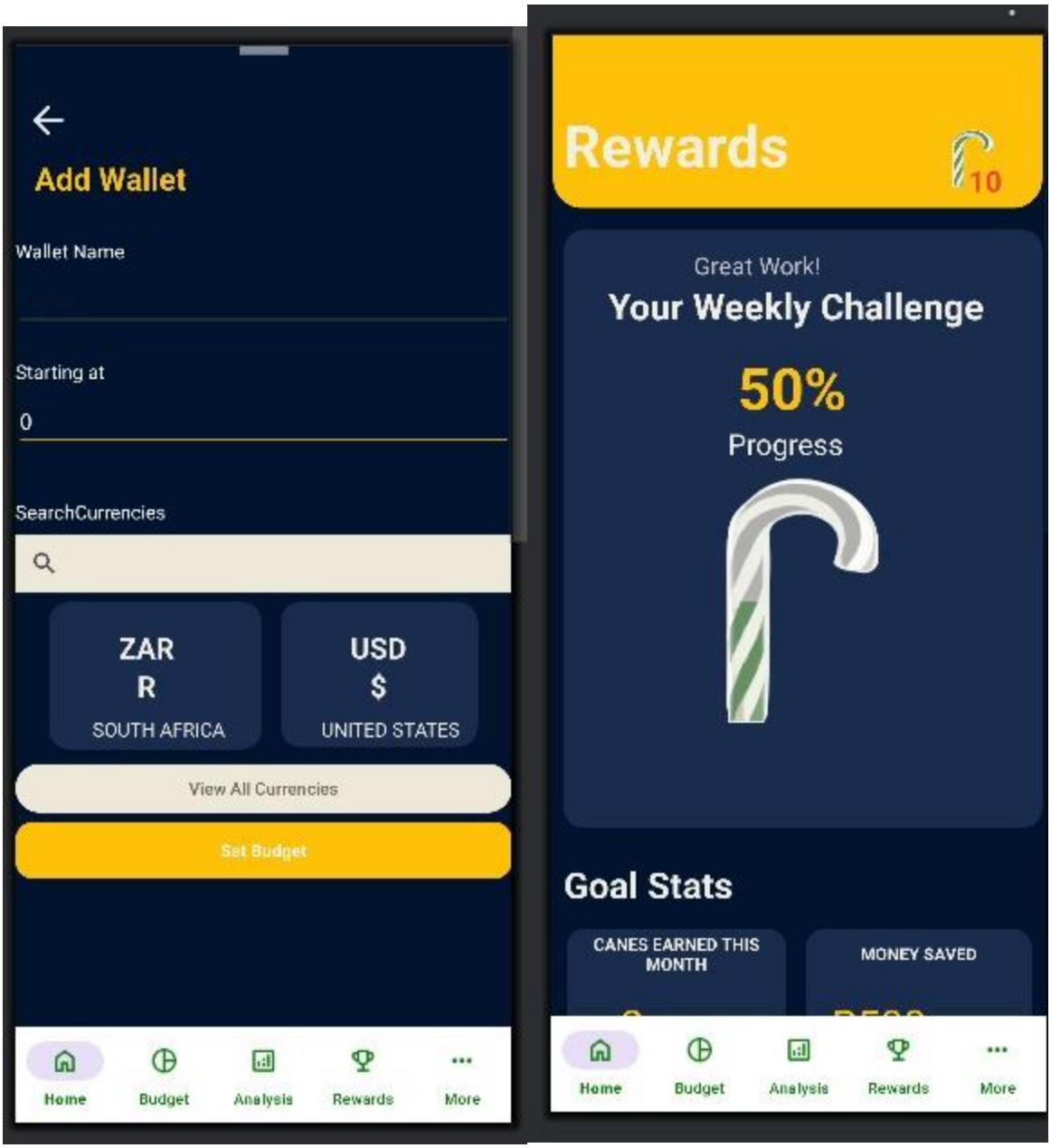
Figure 18: Home Screen Dark mode



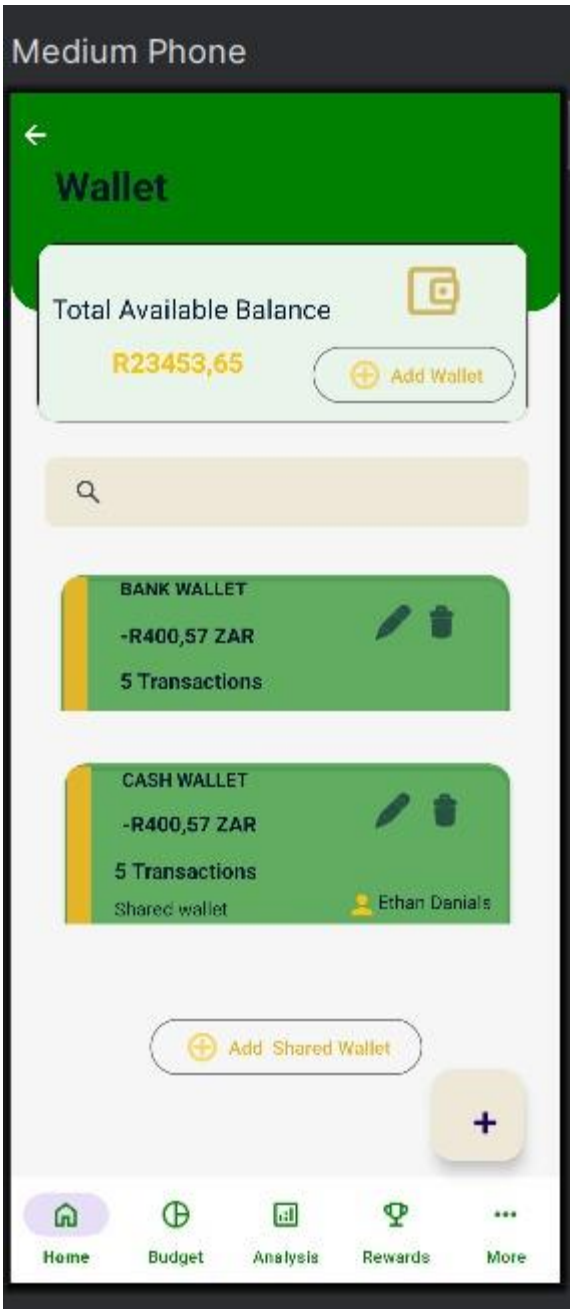
Figure 19:Home screen light mode

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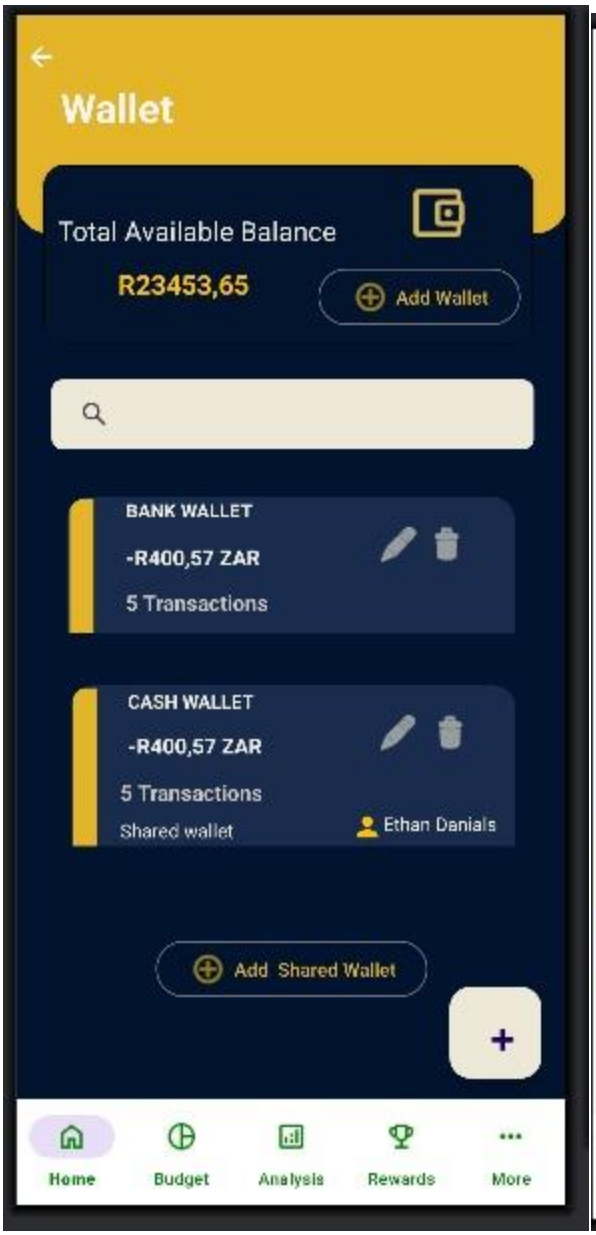


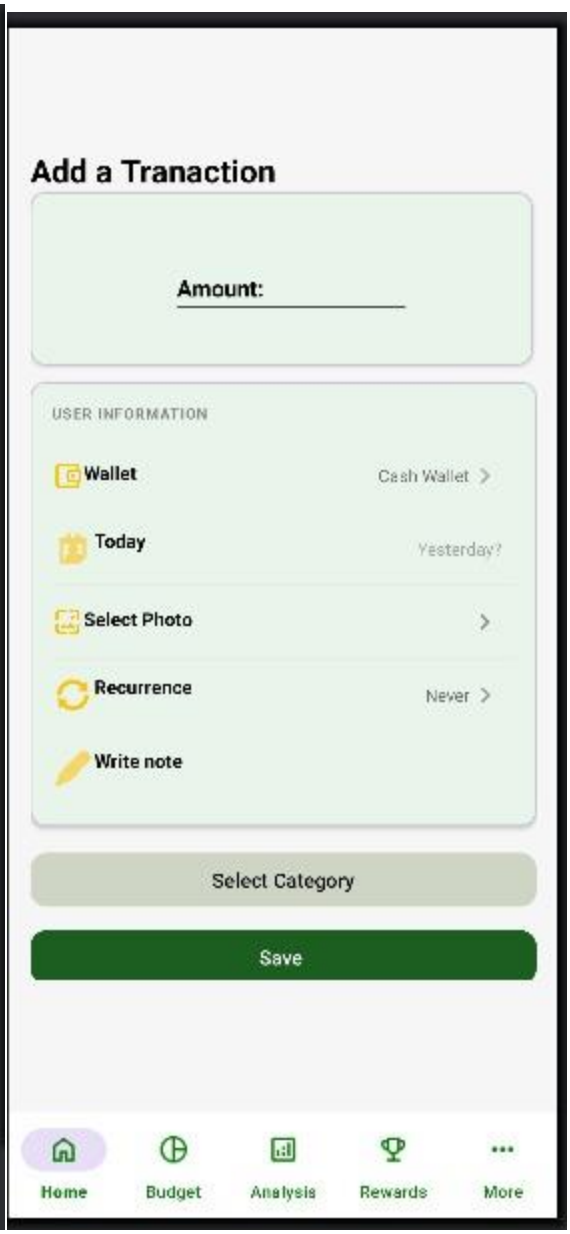

Add Wallet
Wallet Name
Starting at
0
SearchCurrencies
ZAR
R
SOUTH AFRICA
USD
$
UNITED STATES
View All Currencies
Set Budget
Home
Budget
Analysis
Rewards
More
Rewards
10
Great Work!
Your Weekly Challenge
50%
Progress
Goal Stats
CANES EARNED THIS MONTH
MONEY SAVED
Home
Budget
Analysis
Rewards
More

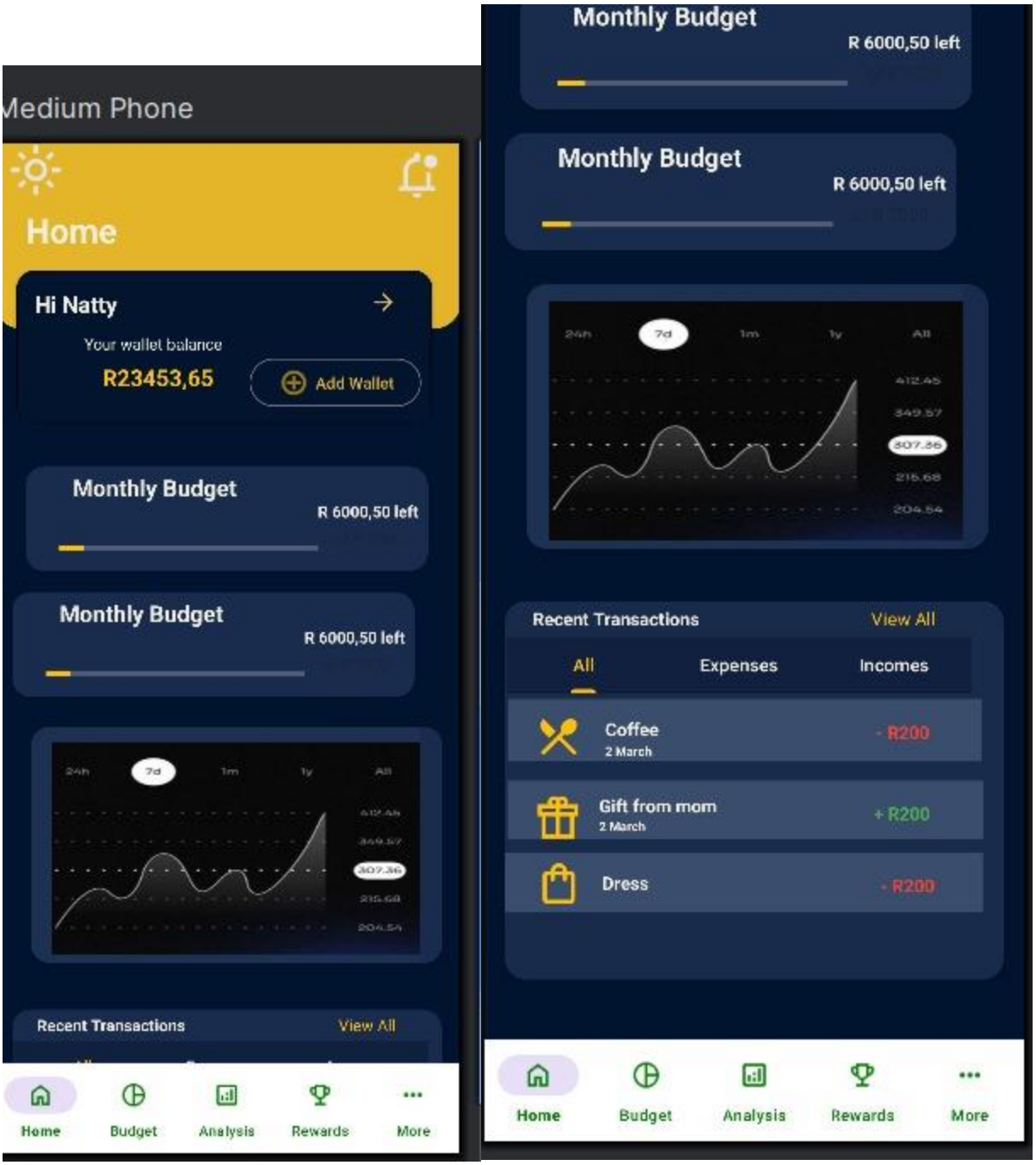
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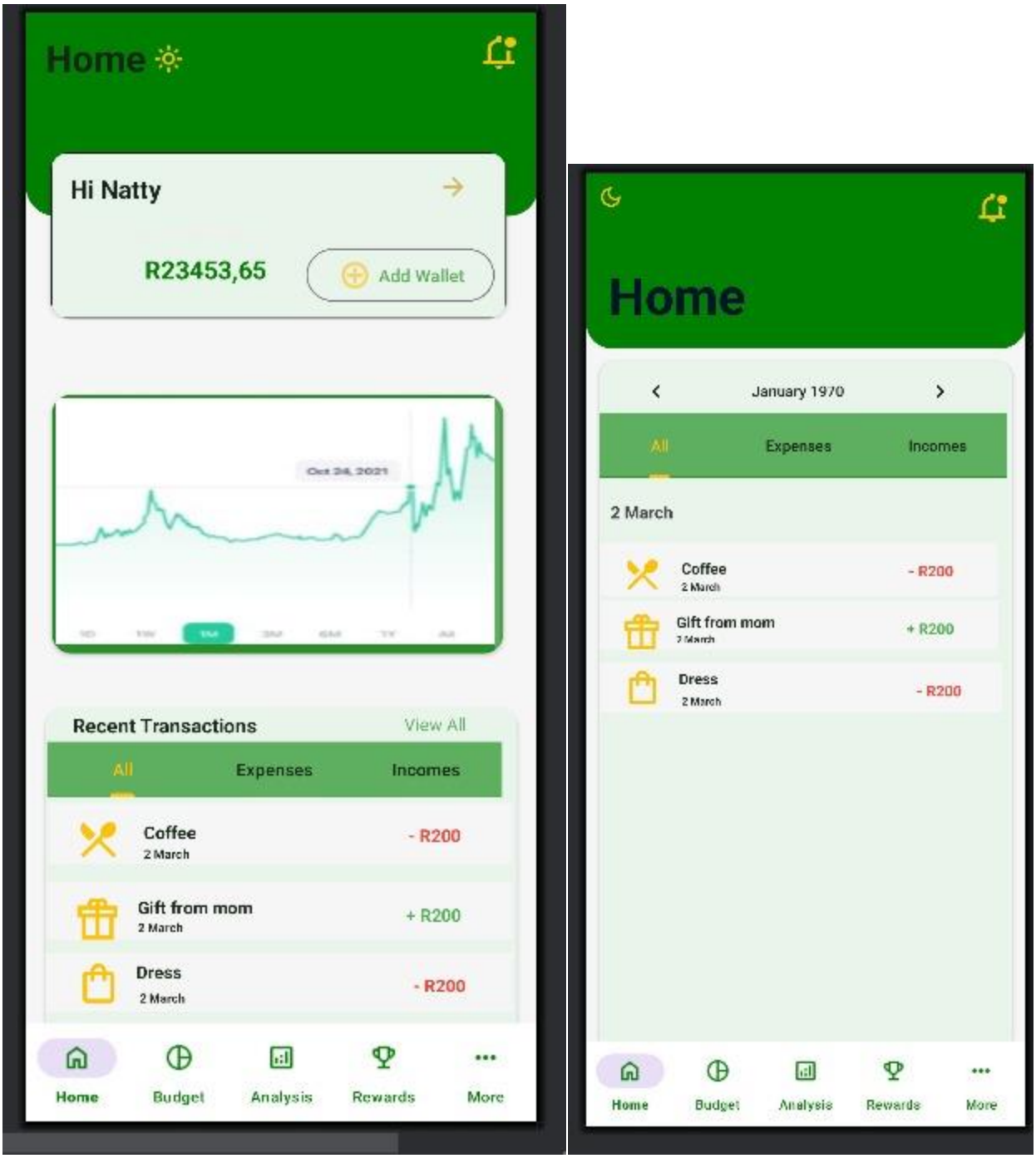



Wallet
Total Available Balance
R23453,65
Add Wallet
BANK WALLET
-R400,57 ZAR
5 Transactions
CASH WALLET
-R400,57 ZAR
5 Transactions
Shared wallet
Ethan Daniels
Add Shared Wallet
Home
Budget
Analysis
Rewards
More

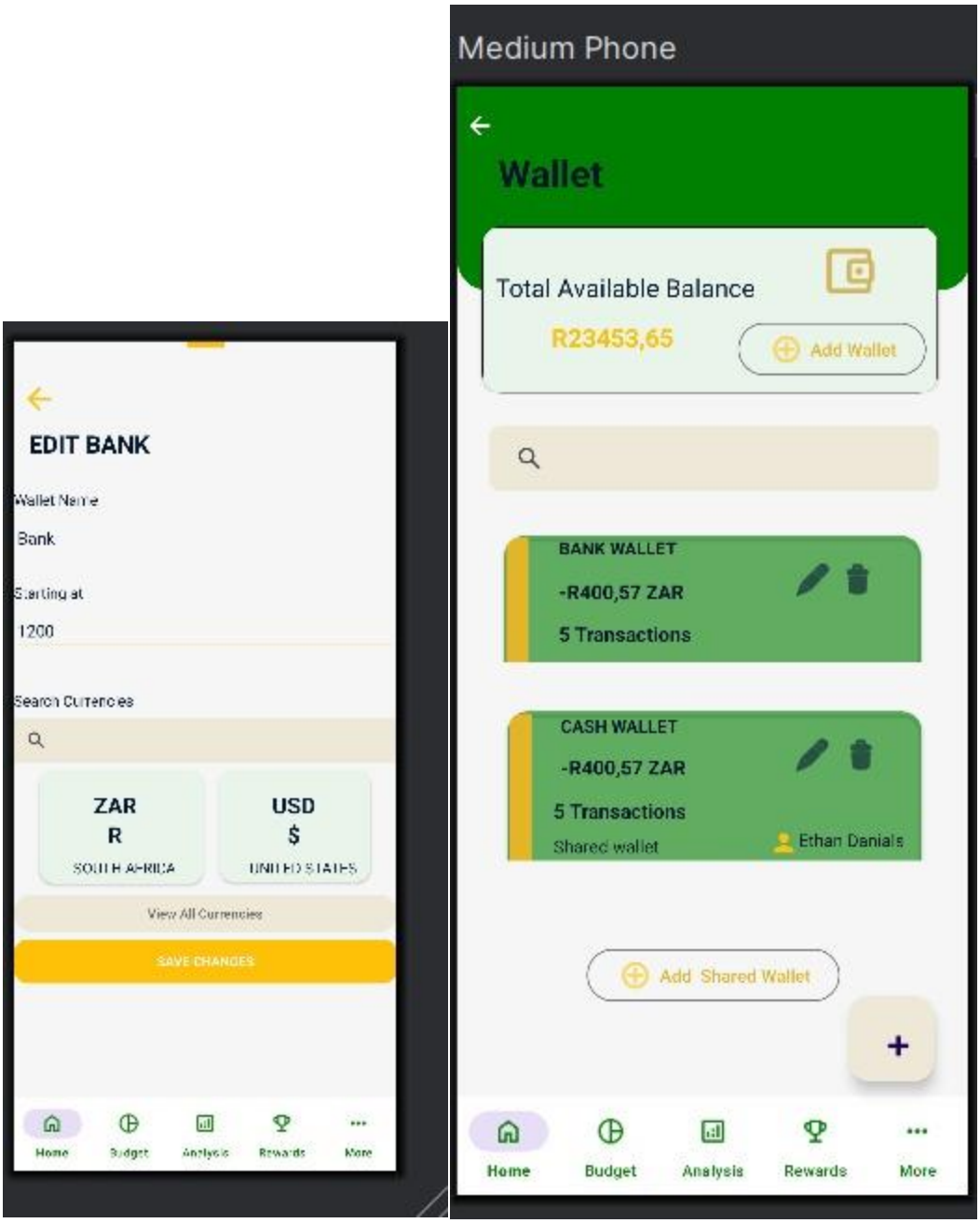

Add a Tranaction
Amount:
USER INFORMATION
Wallet
Cash Wallet
Today
Yesterday?
Select Photo
Recurrence
Never
Write note
Select Category
Save
Home
Budget
Analysis
Rewards
More

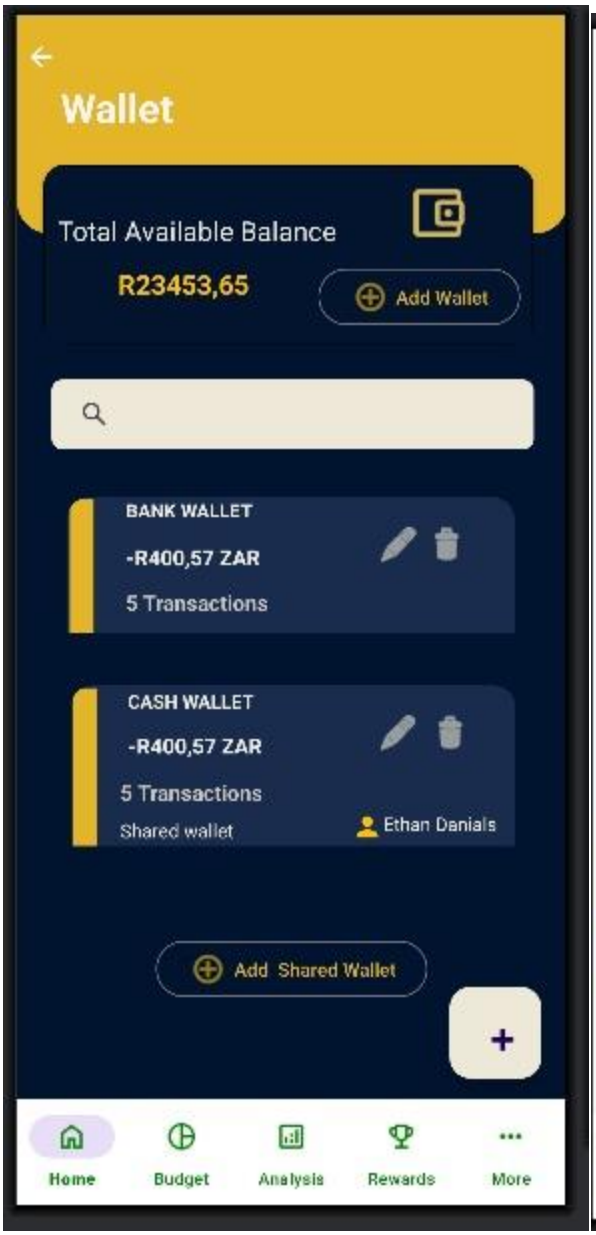

Medium Phone
Home
Hi Natty
Your wallet balance
R23453,65
Add Wallet
Monthly Budget
R 6000,50 left
Monthly Budget
R 6000,50 left
Recent Transactions
View All
Home
Budget
Analysis
Rewards
More
Monthly Budget
R 6000,50 left
Monthly Budget
R 6000,50 left
Recent Transactions
View All
All
Expenses
Incomes
Coffee
2 March
- R200
Gift from mom
2 March
+ R200
Dress
- R200
Home
Budget
Analysis
Rewards
More

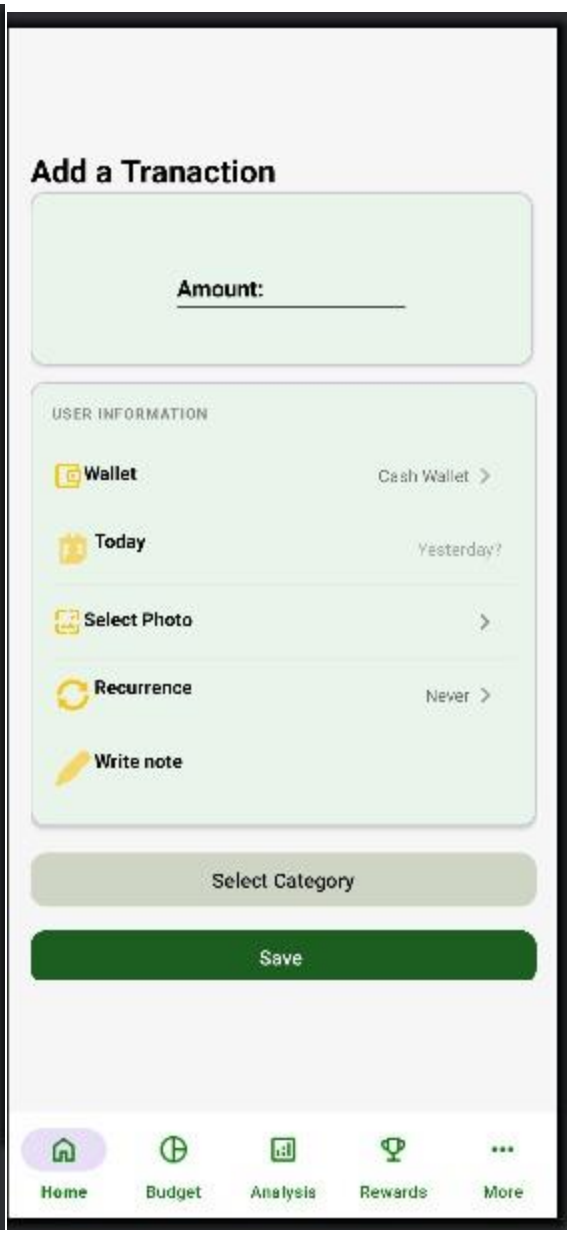
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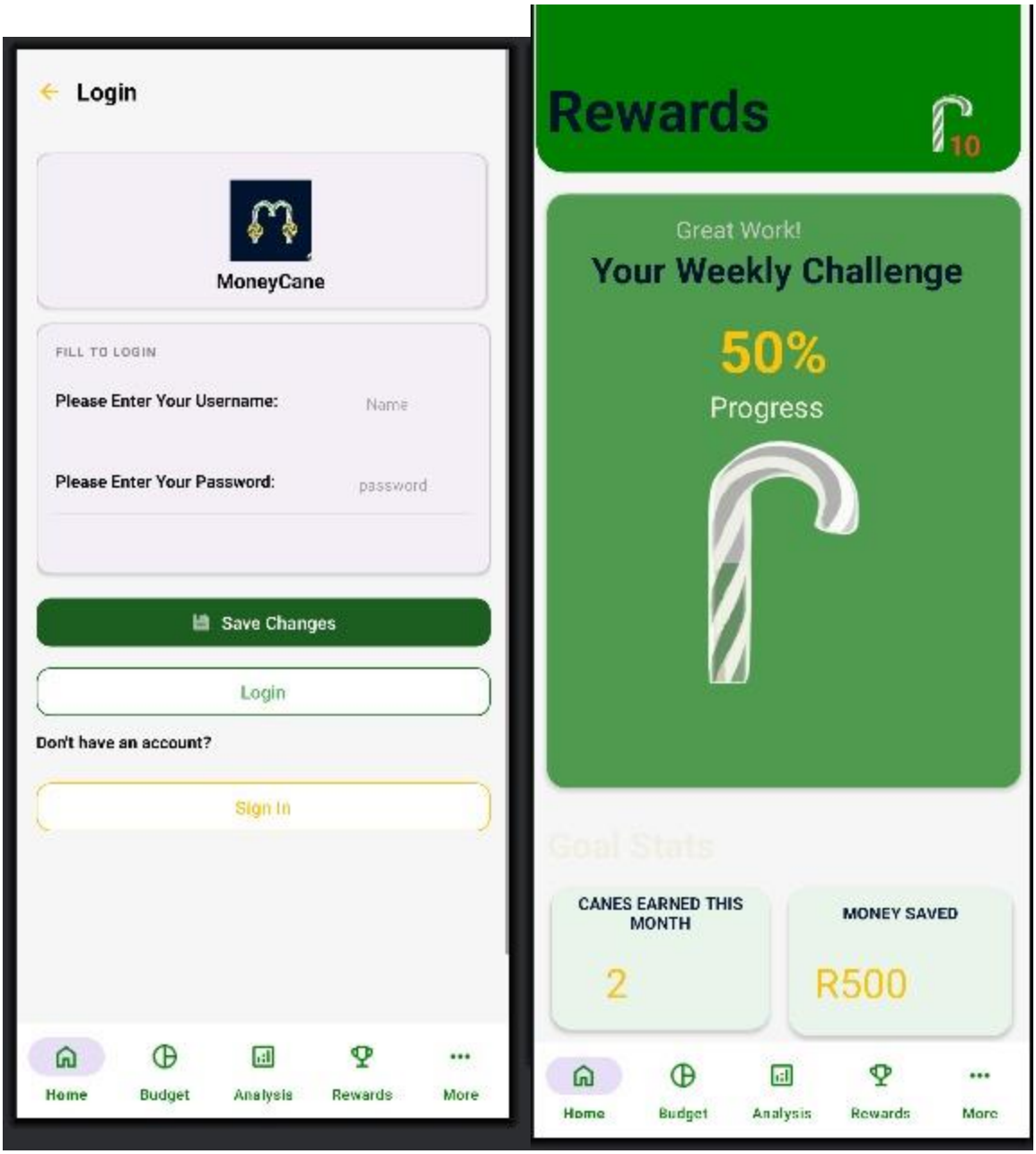


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Wallet
Total Available Balance
R23453,65
Add Wallet
BANK WALLET
-R400,57 ZAR
5 Transactions
CASH WALLET
-R400,57 ZAR
5 Transactions
Shared wallet
Ethan Daniels
Add Shared Wallet
Home
Budget
Analysis
Rewards
More


Add a Tranaction
Amount:
USER INFORMATION
Wallet
Cash Wallet
Today
Yesterday?
Select Photo
Recurrence
Never
Write note
Select Category
Save
Home
Budget
Analysis
Rewards
More


Login
MoneyCane
FILL TO LOGIN
Please Enter Your Username:
Name
Please Enter Your Password:
password
Save Changes
Login
Don't have an account?
Sign In
Home
Budget
Analysis
Rewards
More
Rewards
10
Great Work!
Your Weekly Challenge
50%
Progress
Goal Stats
CANES EARNED THIS MONTH
2
MONEY SAVED
R500
Home
Budget
Analysis
Rewards
More

- Home page where the user can view their wallet balance, their monthly budget and goal progress. They can also view their transactions and spending overview graph all these are customizable; the user can choose what they want to see on their home page
- Rewards page where user views the progress of their weekly goal
- Login page where user enters authentication information to access their account
- Create Transaction Allows user to add transaction, user must add their transaction information.
- User can edit wallet information
- User can add a new wallet, they can put in wallet amount, title, and currency
- Displays all the wallets available on the profile, allows user to add, delete and edit wallets. Special feature to share wallet with other user available.

Navigation Diagram

Please click the YouTube video link to see a presentation of our diagram. The video navigates to each screen and explains the functionality each screen performs.

https://youtu.be/OdplkuSiakw?si=75pUV9oCsIn7-iqV

Gantt Chart

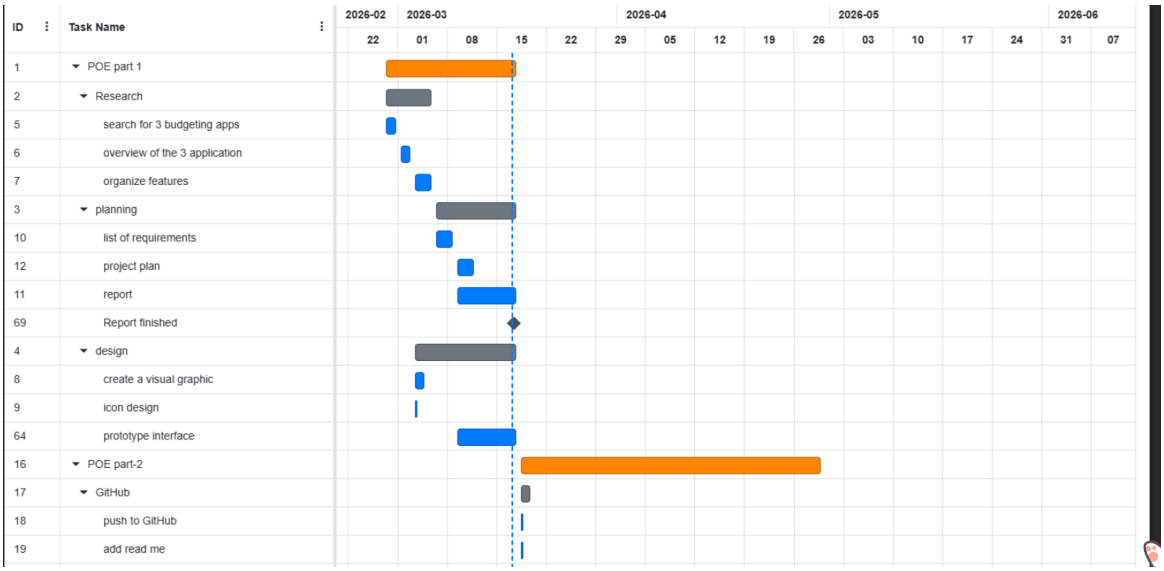


Figure 6: Gantt Chart Part 1



Figure 7:Gantt Chart Part 2

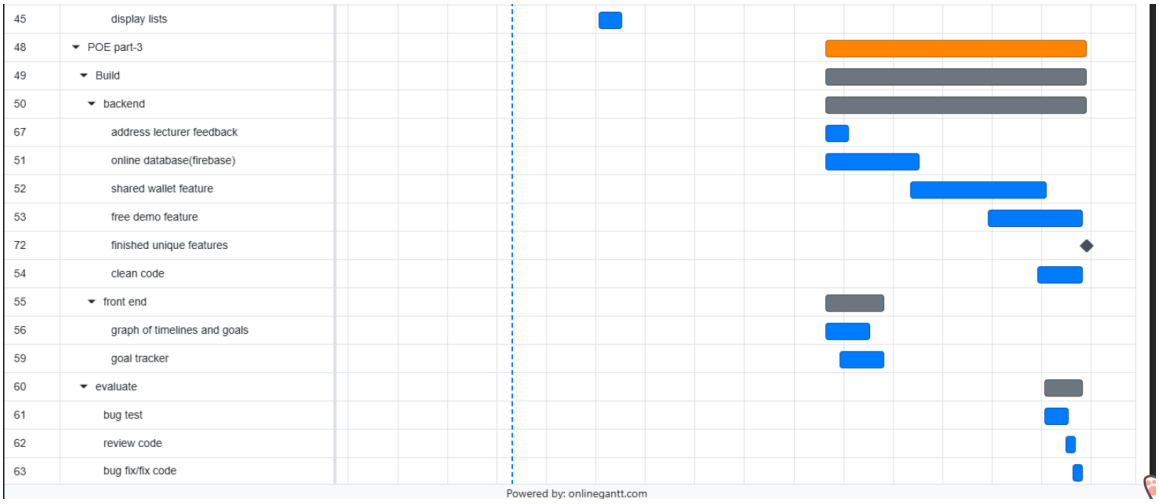


Figure 8: Gantt Chart Part 3

Conclusion

This document has discussed the planning and design of MoneyCane by providing a detailed description of the features and requirements. The Gantt chart has clearly displayed the time frame and highlighted milestones to be achieved. MoneyCane is set to be an app that prioritizes user experience.

References

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